

Last Resort Hotel Management System

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Last Resort Hotels — rapid growth via acquisitions

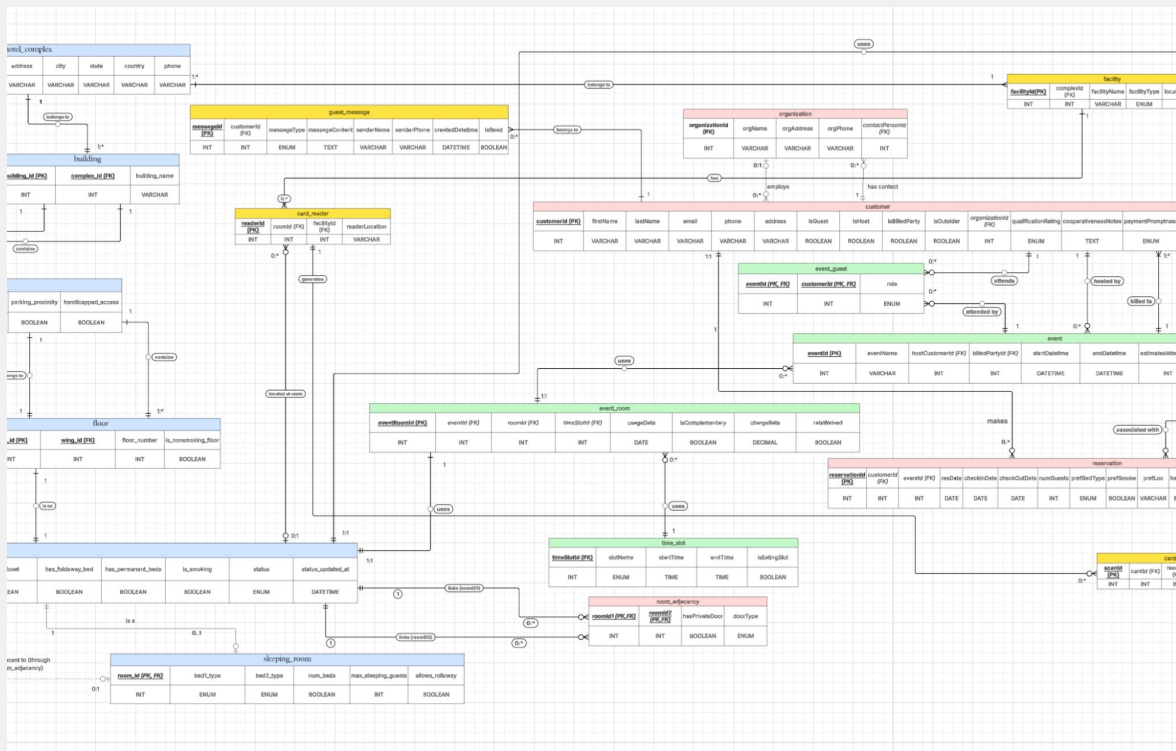
Hundreds of rooms across multiple buildings, wings, floors

Staff can't manually track: room status, reservations, billing, events

Our system solves: room availability, reservation management, event scheduling, charge tracking, guest services

Scenario Overview: Why does Last Resort Hotels need a database system?

ERD Overview ([Link](#))



ERD → Schema Design

Location

hotel_complex → hotel_complex → building → wing → floor
→ room + room_bed, room_adjacency

People

customer, organization

Booking

reservation → room_assignment (prefs vs. actual)
event → event_room + event_guest + time_slot

Billing

charge, billing_split, facility
guest_card → card_reader → card_scan_log
guest_message

23 Tables
in
4 clusters

Design Decisions

1. Combined Room Table (no subtypes)

→ Sleeping + meeting attributes on one table, room_type determines function

2. Roles Through Relationships, Not Flags

→ No is_guest / is_host / is_billed_party columns

3. Reservation ≠ Room Assignment

→ Preferences vs. actual allocation — separated by design

4. Beds as a Separate Table

→ room_bed instead of bed1_type / bed2_type columns

5. Preferences as Key-Value Pairs

→ reservation_preference instead of columns on reservation

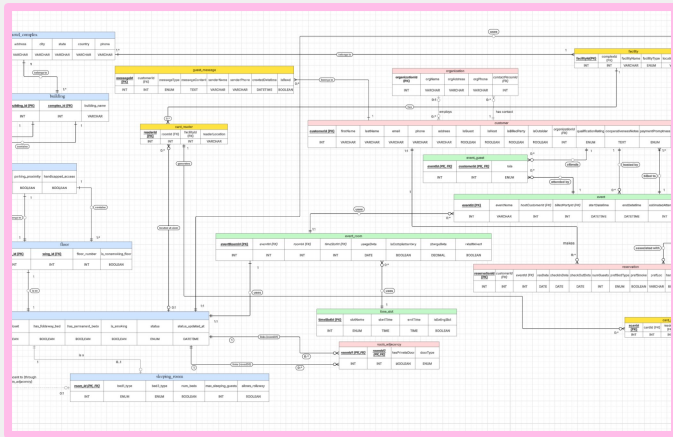
6. Room Adjacency as Self-Referencing M:N

→ room_adjacency junction table with CHECK (roomId1 < roomId2)

Schema Design (Step 1)

ERD → Schema Implementation

- Relational database: SQLite
- Tables created in tables.sql
- 23 tables total
- Core entities: hotel_complex, building, wing, floor, room, customer, reservation, charge



```
27 -- Hotel Complex
28 CREATE TABLE hotel_complex (
29     complexId INTEGER PRIMARY KEY,
30     complexName TEXT NOT NULL,
31     address TEXT NOT NULL,
32     city TEXT NOT NULL,
33     state TEXT NOT NULL,
34     country TEXT NOT NULL,
35     phone TEXT
36 );
37
38 -- Building
39 CREATE TABLE building (
40     buildingId INTEGER PRIMARY KEY,
41     complexId INTEGER NOT NULL,
42     buildingName TEXT NOT NULL,
43     FOREIGN KEY (complexId) REFERENCES hotel_complex(complexId)
44 );
45
```

Schema Design (Step 2)

fillTable.sql inserts all structural and operational data

- Hotel: Last Resort Miami, 100 Ocean Dr, Miami FL
 - 3 buildings: Ocean Tower, Palm Wing, Coral Building
 - 3 wings, 3 floors, 3 rooms (Sleeping, Meeting, Suite)
 - 7 customers, 3 organizations, 3 events, 3 reservations
 - Charges, billing splits, facilities, guest cards, messages

```
1 -- hotel_complex
2 INSERT INTO hotel_complex VALUES (1, 'Last Resort Miami', '100 Ocean Dr', 'Miami', 'FL', 'USA', '305-555-0001');
3
4 -- building
5 INSERT INTO building VALUES (1, 1, 'Ocean Tower');
6 INSERT INTO building VALUES (2, 1, 'Palm Wing');
7 INSERT INTO building VALUES (3, 1, 'Coral Building');
8
9 -- wing
10 INSERT INTO wing VALUES (1, 1, 'A', 1, 1, 0, 1);
11 INSERT INTO wing VALUES (2, 1, 'B', 2, 0, 1, 0);
12 INSERT INTO wing VALUES (3, 2, 'A', 1, 1, 1, 1);
13
14 -- floor
15 INSERT INTO floor VALUES (1, 1, 1, 0);
16 INSERT INTO floor VALUES (2, 1, 2, 1);
17 INSERT INTO floor VALUES (3, 2, 1, 0);
18
19 -- room
20 INSERT INTO room VALUES (1, 1, 1, 101, 'SLEEPING', 250, 1, 1, 1, 1, 1, 0, 1, 0, 'AVAILABLE', '2026-03-30 08:00:00',
21 INSERT INTO room VALUES (2, 1, 2, 201, 'MEETING', 500, 1, 0, 1, 0, 0, 0, 0, 0, 'AVAILABLE', '2026-03-30 08:00:00',
22 INSERT INTO room VALUES (3, 2, 3, 102, 'SUITE', 400, 1, 1, 1, 1, 1, 1, 0, 0, 'OCCUPIED', '2026-03-31 14:00:00', 2,
23
24 -- room_bed
25 INSERT INTO room_bed VALUES (1, 1, 'KING', 'DOUBLE');
26 INSERT INTO room_bed VALUES (2, 1, 'QUEEN', 'DOUBLE');
27 INSERT INTO room_bed VALUES (3, 3, 'KING', 'DOUBLE');
28
```

Queries

Purposes

- 13 queries total
- Daily operations + management reporting
- All queries use JOINS across multiple tables
- WHERE clauses filter for specific conditions
- ORDER BY to surface most relevant results

How we structure queries :

```
10
11 --Q2: Rooms available right now
12 SELECT w.wingDesignation, r.roomType, r.roomNumber, r.baseRentalRate
13 FROM room r
14 JOIN wing w ON r.wingId = w.wingId
15 WHERE r.status = 'AVAILABLE'
16 AND r.roomId NOT IN (
17     SELECT roomId FROM room_assignment WHERE isActive = 1
18 )
19 ORDER BY w.wingSequenceNumber, r.roomNumber
20
```

```
21 --Q3: Total Charges per customer, sorted by highest spend
22 SELECT c.customerId, c.firstName || ' ' || c.lastName AS name,
23 COUNT(ch.chargeId) AS numCharges,
24 SUM(ch.amount) AS totalSpend
25 FROM costumer c
26 JOIN charge ch ON c.costumerId = ch.billedPartyId
27 GROUP BY c.costumerId
28 ORDER BY totalSpend DESC;
29
```

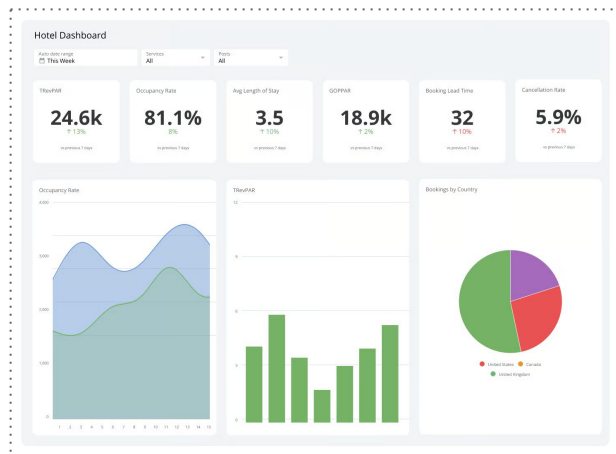
```
74 --Q9: Available rooms with amenities for a given date range
75 -- Replace '2026-04-15' and '2026-04-20' with target dates
76 SELECT r.roomNumber, r.roomType, r.baseRentalRate,
77     r.hasTV, r.hasPhone, r.hasKitchenette,
78     r.isSmoking, r.isHandicapped,
79     w.wingDesignation
80 FROM room r
81 JOIN wing w ON r.wingId = w.wingId
82 WHERE r.status = 'AVAILABLE'
83     AND r.roomType = 'SLEEPING'
84     AND r.roomId NOT IN (
85         SELECT ra.roomId FROM room_assignment ra
86         JOIN reservation res ON ra.reservationId = res.reservationId
87         WHERE res.checkInDate < '2026-04-20'
88             AND res.checkOutDate > '2026-04-15'
89     )
90 ORDER BY r.baseRentalRate;
```


Future Direction

1

More Visualizations

- Graphical dashboard to track revenue growth and guest loyalty



2

More User Input Features

- Room Change Feature
 - Use a `Room_Assignments` table with `reservation_id`, `room_id`, `start_date`, and `end_date`
- Delete guest function

The Room Change Feature form allows users to update room assignments. It is divided into two main sections: 'Current Room' and 'New Room', connected by a right-pointing arrow.

Current Room:

- Room Number: 412 - Deluxe King
- Building / Floor: A / 4
- Status: OCCUPIED
- Rate per Night: \$200.00

New Room:

- Room Number *: 518 - Deluxe King
- Building / Floor: A / 5
- Rate per Night: \$200.00
- Change Date *: 06/03/2025
- Change Time: After 12:00 PM (Checkout time)
- Reason (Optional): Guest requested higher floor

Demo

Last Resort Hotels
RESERVATIONS & OPERATIONS

Search guests or rooms

Search

Reports ▾

+ Customer

+ Reservation

New Customer

Register a new guest, host, or billing contact.

FIRST NAME *

LAST NAME *

EMAIL *

PHONE *

ADDRESS
Street, City, State

ORGANIZATION
— none — ▾

QUALIFICATION RATING
NEW ▾

PAYMENT PROMPTNESS
FAIR ▾

☐ Keep guest location confidential

Create customerCancel

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Last Resort Hotels
RESERVATIONS & OPERATIONS

Search guests or rooms

Search

Reports ▾

+ Customer

+ Reservation

New Reservation

Reserve sleeping rooms for a guest. Specific rooms get assigned closer to the check-in date by the front-desk staff.

GUEST *
— select a customer — ▾

LINKED EVENT (OPTIONAL)
— none — ▾

CHECK-IN DATE *
mm/dd/yyyy 📅

CHECK-OUT DATE *
mm/dd/yyyy 📅

NUMBER OF GUESTS
1

STATUS
PENDING ▾

DEPOSIT AMOUNT (\$)

☐ Deposit required☐ Deposit already paid

Create reservationCancel

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Thank you!